



Pearson New International Edition

Excursions in Modern Mathematics

Peter Tannenbaum

Eighth Edition



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The Mathematics of Apportionment

Making the Rounds

April 1, 2010 was not just any ordinary April Fool's Day. With an exquisite sense of irony, the United States Census Bureau designated April 1 as "Census Day" for the 2010 Census—the day when the population of the United States was officially counted. And indeed it was—there seemed to be around 308,745,538 of us going about our business that day

U.S. Census Bureau

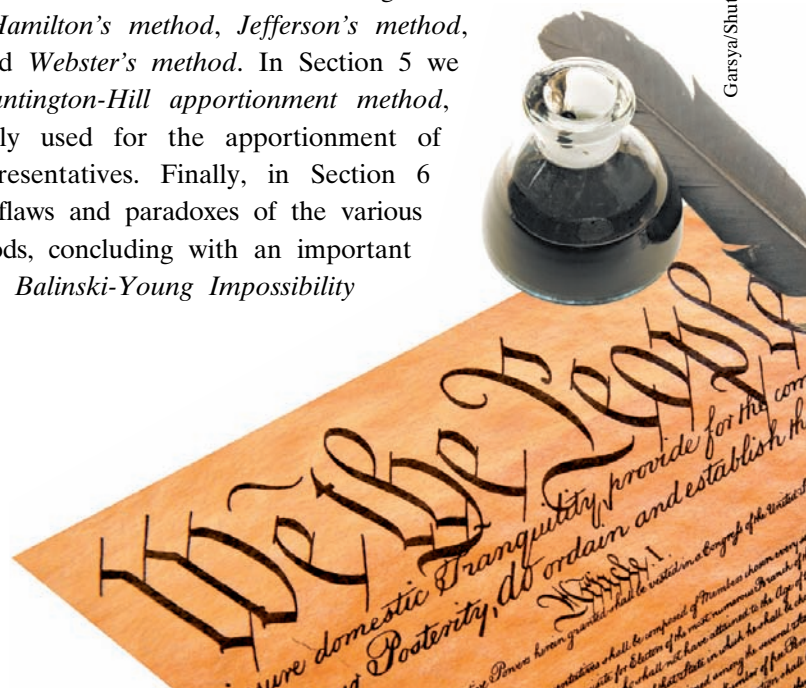


“Representatives . . . shall be apportioned among the several States . . . according to their respective Numbers. The actual Enumeration shall be made . . . every ten Years . . .”

– Article I, Section 2,
U.S. Constitution

The Constitution mandates that every 10 years—on years ending with a 0—the government produce a “head count” of the U.S. population, broken down by state. (The modern version of the U.S. Census does a lot more than count heads, but at the moment we will focus on just the state population.) The key purpose of the state population numbers is to meet the constitutional requirement of “proportional representation.” The Constitution requires that seats in the U.S. House of Representatives be “apportioned among the . . . States according to their respective Numbers,” so every 10 years two things must happen: (a) the state populations must be determined (that’s the Census part), and (b) the seats in the House of Representatives must be apportioned to the states based on their populations (that’s the “apportionment” part). It doesn’t sound all that complicated, does it? You will be surprised!

This chapter is an excursion into the mathematics behind the apportionment part, and it contains many interesting twists and turns. There are also a lot of parallels between apportionment and voting. We will start the chapter with an introduction to the basic concepts of apportionment: What is an *apportionment problem*? What are the elements common to all apportionment problems? What is an *apportionment method*? In Sections 2 through 4 we will introduce four classic apportionment methods named after four famous figures in American history: *Hamilton’s method*, *Jefferson’s method*, *Adams’s method*, and *Webster’s method*. In Section 5 we will discuss the *Huntington-Hill apportionment method*, the method currently used for the apportionment of the House of Representatives. Finally, in Section 6 we will discuss the flaws and paradoxes of the various apportionment methods, concluding with an important result known as the *Balinski-Young Impossibility Theorem*.





Apportionment Problems and Apportionment Methods

“ap·por·tion: to *divide* and *share* out according to a *plan*; to make a *proportionate* division or distribution of . . .”

– Merriam-Webster Dictionary*

Obviously, the verb *apportion* is the key word in this chapter. There are two critical elements in the dictionary definition of the word: (1) We are *dividing* and *sharing* things, and (2) we are doing this on a *proportional* basis and in a *planned*, organized fashion.

We will start this section with a pair of examples that illustrate the nature of the problem we are dealing with. These examples will raise several questions, but the answers will come later.

EXAMPLE 1 KITCHEN CAPITALISM

Mom has a total of 50 identical pieces of candy (let’s say caramels), which she is planning to divide among her five children (this is the “divide and share” part). Like any good mom, she is intent on doing this fairly. Of course, the easiest thing to do would be to give each child 10 caramels—by most standards, that would be fair. Mom, however, is thinking of the long-term picture—she wants to teach her children about the value of work and about the relationship between work and reward. This leads her to the following idea: She announces to the kids that the candy is going to be divided at the end of the week in proportion to the amount of time each of them spends helping with the weekly kitchen chores—if you worked twice as long as your brother you get twice as much candy, and so on (this is the “proportionate” division part). Unwittingly, mom has turned this division problem into an apportionment problem.

At the end of the week, the numbers are in. Table 1 shows the amount of work done by each child during the week. (Yes, mom did keep up-to-the-minute records!)

Child	Alan	Betty	Connie	Doug	Ellie	Total
Minutes worked	150	78	173	204	295	900

■ TABLE 1 Work (in minutes) per child

According to the ground rules, Alan, who worked 150 out of a total of 900 minutes, should get $8\frac{1}{3}$ pieces. (Setting up the proportion $\frac{150}{900} = \frac{x}{50}$ and solving for x gives $x = 8\frac{1}{3}$.) Here comes the problem: Since the pieces of candy are indivisible, it is impossible for Alan to get his $8\frac{1}{3}$ pieces—he can get 8 pieces (and get shorted) or he can get 9 pieces (and someone else will get shorted). A similar problem occurs with each of the other children. Betty’s exact fair share should be $4\frac{1}{3}$ pieces; Connie’s should be $9\frac{11}{18}$ pieces; Doug’s, $11\frac{1}{3}$ pieces; and Ellie’s, $16\frac{7}{18}$ pieces. (Be sure to double-check these figures!) Because none of these shares can be realized, an absolutely fair apportionment of the candy is going to be impossible. What should mom do?

Example 1 shows all the elements of an apportionment problem—there are objects to be divided (the pieces of candy), and there is a proportionality criterion for the division (number of minutes worked during the week). We will say that the pieces of candy are *apportioned* to the kids, and we will describe the final solution as an *apportionment* (Alan’s *apportionment* is x pieces, Betty’s *apportionment* is y pieces, etc.).

Let’s now consider a seemingly different apportionment problem.

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EXAMPLE 2 THE INTERGALACTIC CONGRESS OF UTOPIA

It is the year 2525, and all the planets in the Utopia galaxy have finally signed a peace treaty. Five of the planets (Alanos, Betta, Conii, Dugos, and Ellisium) decide to join forces and form an Intergalactic Federation. The Federation will be ruled by an Intergalactic Congress consisting of 50 “elders,” and the 50 seats in the Intergalactic Congress are to be *apportioned* among the planets according to their respective populations.

The population data for each of the planets (in billions) are shown in Table 2. Based on these population figures, what is the correct *apportionment* of seats to each planet?

Planet	Alanos	Betta	Conii	Dugos	Ellisium	Total
Population	150	78	173	204	295	900

Source: Intergalactic Census Bureau.

■ **TABLE 2** Intergalactic Federation: Population figures (in billions) for 2525

Example 2 is another example of an apportionment problem—here the objects being divided are the 50 seats in the Intergalactic Congress, and the proportionality criterion is population. But there is more to it than that. The observant reader may have noticed that the numbers in Examples 1 and 2 are identical—it is only the framing of the problem that has changed. Mathematically speaking, Examples 1 and 2 are one and the same apportionment problem—solve either one and you have solved both!

Apportionment: Basic Concepts and Terminology

Between the extremes of apportioning the seats in the Intergalactic Congress of Utopia (important, but too far away!) and apportioning the candy among the children in the house (closer to home, but the galaxy will not come to an end if the apportionments aren’t done right!) fall many other, real-life problems that are both important and relevant: assigning classrooms to departments in a university, assigning nurses to shifts in a hospital, etc. But the gold standard for apportionment applications is the allocation of seats to “states” in a legislature, and, thus, it is standard practice to borrow the terminology of legislative apportionment and apply it to apportionment problems in general.

The basic elements of every apportionment problem are as follows:

- **The “states.”** This is the term we will use to describe the parties having a stake in the apportionment. Unless they have specific names (Alanos, Betta, etc.), we will use A , B , C , etc. to denote the states. Note that the “states” can be districts, counties, planets, and even people.
- **The “seats.”** This term describes the set of M *identical, indivisible objects* that are being divided among the N states. For convenience, we will assume that there are more seats than there are states, thus ensuring that every state can potentially get a seat. (This assumption does not imply that every state *must* get a seat! Such a requirement is not part of the general apportionment problem, although it is part of the constitutional requirements for the apportionment of the U.S. House of Representatives.)
- **The “populations.”** This is a set of N positive numbers (for simplicity we will assume that they are whole numbers) that are used as the basis for the apportionment of the seats to the states. We will use $p_1, p_2, \dots, p_N$ to denote the states’ respective populations and P to denote the total population ($P = p_1 + p_2 + \dots + p_N$).

- **An apportionment method.** A systematic procedure that *guarantees* a division of the M seats (no more or no less) to the N states using some formula that is based on the state populations.

We now formally define several important concepts that we will use throughout this chapter.

- **The standard divisor (SD).** This is the ratio of population to seats ($SD = \frac{P}{M}$). It gives us a unit of measurement (SD people = 1 seat) for our apportionment calculations.
- **The standard quotas.** The standard quota of a state (sometimes called the *fair quota*) is the exact fractional number of seats that the state would get if fractional seats were allowed. We will use the notation $q_1, q_2, \dots, q_N$ to denote the standard quotas of the respective states. To find a state's standard quota, we divide the state's population by the standard divisor (for a state with population p , its standard quota is $q = \frac{p}{SD}$). In general, the standard quotas can be expressed as decimals or fractions—it would be almost a miracle if one of them turned out to be a whole number—and when using the decimal form, it's customary to round them to two or three decimal places.
- **Upper and lower quotas.** Associated with each standard quota are two other important numbers—the **lower quota** (the standard quota rounded down) and the **upper quota** (the standard quota rounded up). In the unlikely event that the standard quota is a whole number, the lower and upper quotas are the same. We will use L 's to denote lower quotas and U 's to denote upper quotas. For example, the standard quota $q = 32.92$ has lower quota $L = 32$ and upper quota $U = 33$.

We will now introduce the most important apportionment example of this chapter. We will return to this example many times. While the story behind the example is obviously fiction, the issues it raises are very real.

EXAMPLE 3 THE CONGRESS OF PARADOR

Parador is a small republic located in Central America and consists of six states: Azucar, Bahia, Cafe, Diamante, Esmeralda, and Felicidad (A, B, C, D, E , and F for short). There are 250 seats in the Congress, which, according to the laws of Parador, are to be apportioned among the six states in proportion to their respective populations. What is the “correct” apportionment?

Table 3 shows the population figures for the six states according to the most recent census.

State	A	B	C	D	E	F	Total
Population	1,646,000	6,936,000	154,000	2,091,000	685,000	988,000	12,500,000

■ **TABLE 3** Republic of Parador (Populations by state)

The first step we will take to tackle this apportionment problem is to compute the standard divisor (SD). In the case of Parador, the standard divisor is $SD = \frac{12,500,000}{250} = 50,000$. (Typically, the standard divisor is not going to turn out to be such a nice whole number, but it is always going to be a rational number.)

The standard divisor $SD = 50,000$ tells us that in Parador, each seat in the Congress corresponds to 50,000 people. We can now use this yardstick to find the number of seats that each state *should get by the proportionality criterion*—all we have to do is divide the state's population by 50,000. These are the standard quotas. For example, take state A . If we divide the population of A by 50,000, we get

$q_1 = \frac{1,646,000}{50,000} = 32.92$. If seats in the Congress could be apportioned in fractional parts, then the fair apportionment to *A* would be 32.92 seats, but, of course, this is impossible—seats in the Congress are indivisible objects! Thus, the standard quota of 32.92 is a number that we cannot meet, although in this case we might be able to come close to it: The next best solution might be to apportion 33 seats to state *A*. Hold that thought!

Table 4 shows the standard quotas for each of the states in Parador (rounded to two decimal places). Notice that the sum of the standard quotas equals 250, the number of seats being apportioned.

State	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	Total
Population	1,646,000	6,936,000	154,000	2,091,000	685,000	988,000	12,500,000
Standard quota	32.92	138.72	3.08	41.82	13.70	19.76	250

■ **TABLE 4** Republic of Parador: Standard quotas for each state ($SD = 50,000$)

Standard quotas are the benchmark for a *fair apportionment*, but we must somehow convert these standard quotas to whole numbers. This is where the rubber meets the road: How do we round the quotas into whole numbers?

At first glance, this seems like a dumb question. After all, we all learned in school how to round decimals to whole numbers—round down if the fractional part is less than 0.5, round up otherwise. This kind of rounding is called *rounding to the nearest integer*, or simply *conventional rounding*. Unfortunately, conventional rounding will not work in this example, and Table 5 shows why not—we would end up giving out 251 seats in the Congress, and there are only 250 seats to give out! (Add an extra seat, you say? Sorry, this is not a banquet. The number of seats is fixed—we don’t have the luxury of adding or subtracting seats!)

State	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	Total
Standard quota	32.92	138.72	3.08	41.82	13.70	19.76	250.00
Nearest integer	33	139	3	42	14	20	251

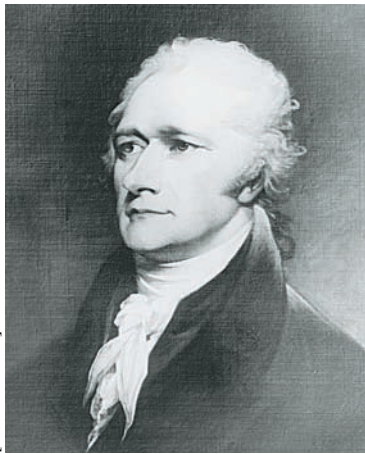
■ **TABLE 5** Conventional rounding of the standard quotas

The most important moral of Example 3 is that conventional rounding of the standard quotas is *not* an apportionment method. It can end up apportioning more than M seats (as in Example 3) or less than M seats. Sometimes things do fall into place, and it ends up apportioning exactly M seats, but since this is not guaranteed, we can’t count on it as an apportionment method. So, what do we try next? For now, we leave this example—a simple apportionment problem looking for a simple solution—with no answers (but, we hope, with some gained insight!). Our search for a good solution to this problem will be the theme of our journey through the rest of this chapter.

2

Hamilton’s Method

Alexander Hamilton first proposed this method in 1792 as a way to apportion the U.S. House of Representatives (for more details see “A Brief History of Apportionment in the United States” in the Conclusion). Hamilton’s method is also known as *Vinton’s method* and *the method of largest remainders*.

Alexander Hamilton
(1757–1804)

Hamilton's method can be described quite briefly: *Every state gets at least its lower quota. As many states as possible get their upper quota, with the one with highest residue (i.e., fractional part) having first priority, the one with second highest residue second priority, and so on.* A little more formally, it goes like this:

■ HAMILTON'S METHOD

- **Step 1.** Calculate each state's standard quota.
- **Step 2.** Round the standard quotas down and give to each state its *lower quota*.
- **Step 3.** Give the surplus seats (one at a time) to the states with the largest *residues* (fractional parts) until there are no more surplus seats.

EXAMPLE 4

PARADOR'S CONGRESS (HAMILTON'S METHOD)

As promised, we are revisiting the Parador Congress apportionment problem. We are now going to find our first real solution to this problem—the solution given by Hamilton's method (we will call it the *Hamilton apportionment*). Table 6 shows all the details. (Reminder to the reader: The standard quotas in the second row of the table were originally computed in Example 3.) After each state receives its lower quota, there are 4 surplus seats left. These go to *A* first (0.92), *D* second (0.82), *F* third (0.76), and *B* last (0.72). *E* and *C* get no extra seats.

State	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	Total
Standard quota	32.92	138.72	3.08	41.82	13.70	19.76	250.00
Lower quota	32	138	3	41	13	19	246
Residue	0.92	0.72	0.08	0.82	0.70	0.76	4.00
Order of Surplus	First	Fourth		Second		Third	
Apportionment	33	139	3	42	13	20	250

■ **TABLE 6** Parador's Congress: the Hamilton apportionment

At first glance, Hamilton's method appears to be a good method for apportioning seats, but a careful look at Table 6 already shows hints of possible unfairness. Compare the fates of state *B*, with a residue of 0.72, and state *E*, with a residue of 0.70. State *B* gets the last surplus seat; state *E* gets nothing! Sure enough, 0.72 is more than 0.70, so following the rules of Hamilton's method, we give priority to *B*. By the same token, *B* is a huge state, and as a percentage of its population the 0.72 represents an insignificant amount, whereas state *E* is a relatively small state, and its residue of 0.70 (for which it gets nothing) represents more than 5% of its population ($\frac{0.70}{13.70} \approx 0.051 = 5.1\%$).

It could be reasonably argued that Hamilton's method has a major flaw in the way it relies entirely on the size of the residues without consideration of what those residues represent as a percent of the state's population. In so doing, Hamilton's method creates a systematic bias in favor of larger states over smaller ones. This is bad—a good apportionment method should be *population neutral*, meaning that it should not be biased in favor of large states over small ones or vice versa.

In Section 6 we will see that bias in favor of large states is only one of the many serious flaws of Hamilton's method.

Our next example gives us a preview of some of the issues that can arise when using Hamilton's method. The numbers used in the example are clearly made up, but the situation the example illustrates is very real.

EXAMPLE 5 THE BAMA PARADOX

The small country of Calavos consists of three states: Bama, Tecos, and Ilnos. With a total population of $P = 200,000$ and $M = 200$ seats in the House of Representatives, we have an easy calculation of the standard divisor ($SD = 1000$). With such a beautiful standard divisor (of course, the numbers are rigged!), we can quickly find the standard quotas (just divide the populations by 1000). Table 7 shows the details of the apportionment of the 200 seats under Hamilton's method: Bama gets 10 seats, Tecos get 90 seats, Ilnos gets 100 seats.

	Bama	Tecos	Ilnos	Total
Population	9400	90,300	100,300	200,000
Standard quota	9.4	90.3	100.3	200
Lower quota	9	90	100	199
Surplus	1	0	0	1
Hamilton apportionment	10	90	100	200

■ **TABLE 7** Hamilton's method with $M = 200$, $SD = 1000$

Now we will ask ourselves what happens if the number of seats is increased to 201, but nothing else changes. Since there is one more seat to give out, the apportionment has to be recomputed. Table 8 shows the new apportionment for a House with 201 seats. First, we need to find the standard divisor. Here $SD = \frac{200,000}{201} \approx 995.02$. The standard quotas (rounded to two decimal places) are, respectively, 9.45, 90.75, and 100.8 (second row of Table 8). We now have two surplus seats, and they go to Ilnos first and Tecos second. Bama's apportionment is now down to 9 seats (Table 8).

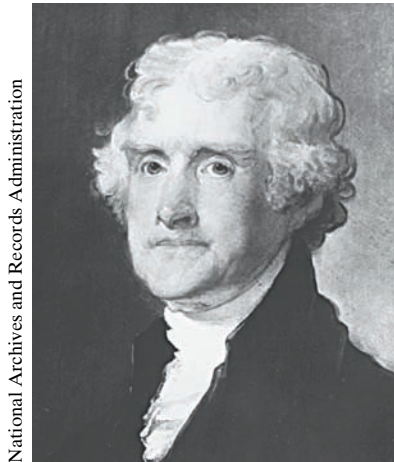
	Bama	Tecos	Ilnos	Total
Population	9400	90,300	100,300	200,000
Standard quota	9.45	90.75	100.8	201
Lower quota	9	90	100	199
Surplus	0	1	1	2
Hamilton apportionment	9	91	101	201

■ **TABLE 8** Hamilton's method with $M = 201$, $SD \approx 995.02$

Strangely, when there were 200 seats to be apportioned, Hamilton's method gave Bama 10 seats, but when the number of seats was increased to 201 Hamilton's method gave Bama 9 seats. How could this happen? Notice the effect that the increase in the number of seats has on the size of the residues: In a House with 200 seats, Bama is at the head of the priority line for surplus seats, but when the number of seats goes up to 201, Bama gets shuffled to the back of the line. Welcome to the wacky arithmetic of Hamilton's method!

3

Jefferson's Method



National Archives and Records Administration

Thomas Jefferson
(1743–1826)

Our next method is attributed to Thomas Jefferson, who proposed it as the right apportionment method for the U.S. House of Representatives in 1792. Jefferson's method, as it is known in the United States, is also known as *d'Hondt's method* and *the method of greatest divisors*. (A more detailed account of the use of Jefferson's method to apportion the U.S. House of Representatives can be found in the section "A Brief History of Apportionment in the United States" in the Conclusion.)

Jefferson's method is based on an approach very different from Hamilton's method, so there is some irony in the fact that we will explain the idea behind Jefferson's method by taking one more look at Hamilton's method.

Recall that under Hamilton's method we start by dividing every state's population by the standard divisor. This gives us the standard quotas. Step 2 is then to round *every* state's quota down. Notice that up to this point Hamilton's method uses a uniform policy for all states—every state is treated in exactly the same way. If you are looking for fairness, this is obviously good! But now comes the bad part (Step 3). We have some leftover seats which we need to distribute, but not enough for every state. Thus, we must choose some states over others for preferential treatment. No matter how fair we try to be about it, there is no getting around the fact that some states get extra seats and others don't.

We saw in Example 5 that it is in the handling of the surplus seats (Step 3) that Hamilton's method runs into trouble. So here is an interesting idea: Let's tweak things so that when the quotas are rounded down, *there are no surplus seats!* (The idea, in other words, is to get rid of Step 3.) But under Hamilton's method there is always going to be at least one surplus seat, so how do we work this bit of magic? The answer is by changing the divisor, which then changes the quotas. The idea is that by using a *smaller* divisor, we make the quotas bigger. If we hit it just right, when we round the slightly bigger quotas down . . . poof! The surplus seats are gone. When we pull that bit of numerical magic, we have in fact implemented *Jefferson's method*.

The best way to get a good feel for the inner workings of Jefferson's method is through an example, so we will do that next.

EXAMPLE 6**PARADOR'S CONGRESS (JEFFERSON'S METHOD)**

State	Population	Standard quota	Lower quota
A	1,646,000	32.92	32
B	6,936,000	138.72	138
C	154,000	3.08	3
D	2,091,000	41.82	41
E	685,000	13.70	13
F	988,000	19.76	19
Total	12,500,000	250	246

■ **TABLE 9** Calculations for Example 6 with the standard divisor $SD = 50,000$

Once again we are going to apportion the 250 seats in Parador's Congress, this time using Jefferson's method. For comparison purposes, we start with a review of how Hamilton's method tackles this problem. The first two steps in Hamilton's method are to compute the standard quotas (by dividing the populations by the standard divisor $SD = 50,000$) and then round them down to the lower quotas. These calculations are shown in Table 9. We know that there is a surplus of 4 seats and that Step 3 of Hamilton's method deals with that issue (but not too well!).

We will now try analogous calculations using a *modified divisor* $d = 49,500$ to divide the populations. This gives us different quotas, which we call *modified quotas*. The second column of Table 10 shows these

modified quotas rounded to two decimal places (using a calculator it's easy to check these numbers). When we round these modified quotas down (third column of Table 10) we end up with 250 seats—no leftovers! This is the apportionment according to Jefferson's method.

State	Population	Modified quota ($d = 49,500$)	Rounded down to
A	1,646,000	33.25	33
B	6,936,000	140.12	140
C	154,000	3.11	3
D	2,091,000	42.24	42
E	685,000	13.84	13
F	988,000	19.96	19
Total	12,500,000		250

■ **TABLE 10** Jefferson's method: Calculations for Example 6 with divisor $d = 49,500$

Two questions come to mind when looking at Example 6: (1) Can we really substitute the standard divisor by some other random divisor?, and (2) if so, where did the divisor $d = 49,500$ come from? We will address each of these questions in order.

The answer to the first question is straightforward: Yes, we can use divisors other than the standard divisor! The standard divisor is a benchmark for a good apportionment, but we are not necessarily bound to it. If the intent is to round the quotas down and use the lower quotas for the apportionments, then the standard divisor is not a good choice because we will have surplus seats to deal with. By choosing the right divisor we can make the problem of surplus seats disappear. One useful way to think about this change in divisors is as *a change in the units of measurement*. Instead of using 1 seat = SD people as our unit, we use 1 seat = d people.

As far as the second question (how did we come up with the divisor $d = 49,500$ in Example 6?), the short answer is that we used educated trial and error. Since our goal is to have no surplus seats, we must make the modified quotas *bigger* than the standard quotas. To do this we must divide the populations by a *smaller* number than the standard divisor. So we start with some number $d < SD$ that is our first guess.

Let's say we make our first guess $d = 49,000$. Table 11 shows the calculation of the modified and lower quotas for this divisor. Using this divisor we see that we overshot our target of 250 by 2 seats. This means that we made our divisor *too small*! So we try another number somewhere between 49,000 and 50,000. A reasonable second guess then is $d = 49,500$. This one we know works! (Fortunately, there is a full range of numbers that work as divisors, so it's not like we have to hit a bullseye. In Example 6 any divisor d between 49,401 and 49,542 also works.)

State	Population	Modified quota ($d = 49,000$)	Rounded down to
A	1,646,000	33.59	33
B	6,936,000	141.55	141
C	154,000	3.14	3
D	2,091,000	42.67	42
E	685,000	13.97	13
F	988,000	20.16	20
Total	12,500,000		252

■ **TABLE 11** Jefferson's method: Calculations for Example 6 with divisor $d = 49,000$

The formal description of Jefferson’s method is surprisingly short. The devil is in the details.

■ JEFFERSON’S METHOD

- **Step 1.** Find a “suitable” divisor d .
- **Step 2.** Using d as the divisor, compute each state’s modified quota (modified quota = state population/ d).
- **Step 3.** Each state is apportioned its modified *lower* quota.

The hard part of implementing Jefferson’s method is Step 1. A flowchart outlining the trial-and-error approach for finding a divisor that works under Jefferson’s method is shown in Fig. 1.

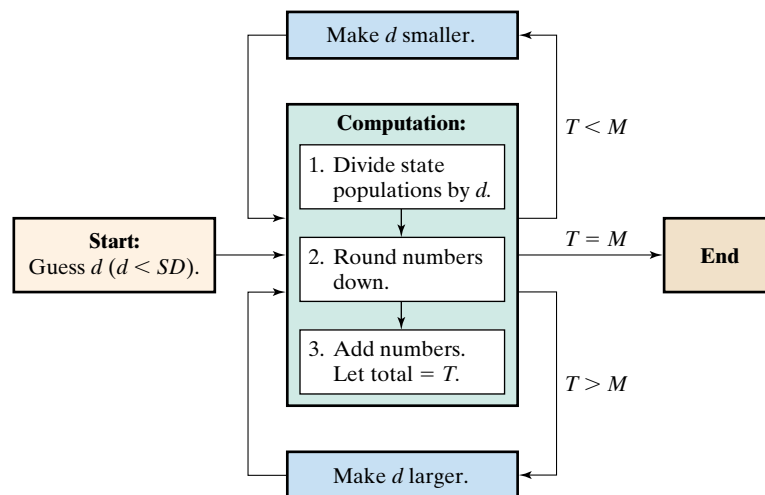


FIGURE 1 Flowchart for trial-and-error implementation of Jefferson’s method.

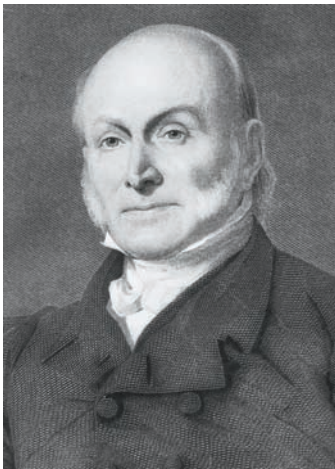
Jefferson’s method is but one of a group of apportionment methods based on the principle that the standard unit (1 seat = SD people) is not an absolute requirement and that, if necessary, we can change to a different unit: 1 seat = d people, where d is a suitably chosen number. The number d is called a **divisor** (sometimes we use the term **modified divisor** to emphasize the point that it is not the standard divisor), and apportionment methods that use modified divisors are called **divisor methods**. Different divisor methods are based on different philosophies of how the **modified quotas should be rounded to whole numbers**, but they all follow one script: When you are done rounding the modified quotas, all M seats have been apportioned (no more, and no less). To be able to do this, you just need to find a suitable divisor d . For Jefferson’s method, the rounding rule is to always round the modified quotas down. In the next two sections we will discuss other divisor methods based on different rounding rules.

4

Adams’s and Webster’s Methods

Adams’s Method

Adams’s method (also known as *the method of smallest divisors*) was first proposed by John Quincy Adams in 1832 as a counterpoint to Jefferson’s method. In essence, Adams’s method is the mirror image of Jefferson’s method: Instead of rounding all quotas down to their lower quotas it rounds them *up to their upper quotas*. This is only possible if we use a modified divisor *larger* than the standard divisor so that the

John Quincy Adams
(1767–1848)

modified quotas are smaller than the standard quotas. As with any divisor method, the trick is to find a divisor that works.

Once again, we will illustrate Adams's method using the apportionment of Parador's Congress.

EXAMPLE 7 PARADOR'S CONGRESS (ADAMS'S METHOD)

Our starting point is to find a suitable divisor d . In Adams's method, the divisor d has to be bigger than the standard divisor $SD = 50,000$. Let's make our first guess $d = 50,500$. Table 12 (third and fourth columns) shows the computations with this divisor. The total number of seats apportioned would be 251. This total is just one seat above our target of 250, so we need to make the quotas just a tad smaller. To do this, we must increase the divisor a little bit.

Let's make our second guess $d = 50,700$. The new computations are shown in the last two columns of Table 12. This divisor works! The apportionment under Adams's method is given in the last column of Table 12. The good news is that there are plenty of other d 's that work just as well (any number between 50,628 and 50,999 will work as a divisor).

State	Population	First guess: $d = 50,500$		Second guess: $d = 50,700$	
		Modified quota	rounded up to	Modified quota	rounded up to
A	1,646,000	32.59	33	32.47	33
B	6,936,000	137.35	138	136.8	137
C	154,000	3.05	4	3.04	4
D	2,091,000	41.41	42	41.24	42
E	685,000	13.56	14	13.51	14
F	988,000	19.56	20	19.49	20
Total	12,500,000		251		250

■ **TABLE 12** Adams's method: Calculations for Example 7 with divisors 50,500 and 50,700

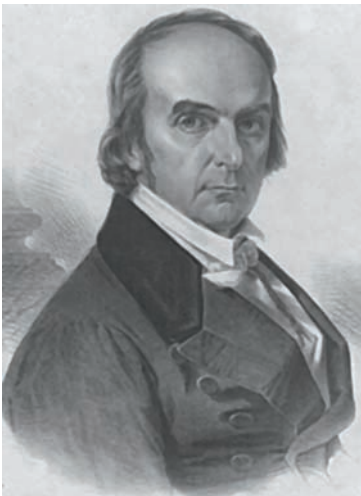
A very short description of Adams's method (remarkably similar to that of Jefferson's method) follows. It looks harmless enough, but Step 1 can take several tries. (Be patient and learn how to get the most out of your calculator.)

■ ADAMS'S METHOD

- **Step 1.** Find a "suitable" divisor d .
- **Step 2.** Using d as the divisor, compute each state's modified quota (modified quota = state population/ d).
- **Step 3.** Each state is apportioned its modified *upper quota*.

Webster's Method

What is the obvious compromise between rounding all the quotas down (Jefferson's method) and rounding all the quotas up (Adams's method)? What about conventional rounding (round the quotas down when the fractional part is less than 0.5 and up otherwise)?



Daniel Webster
(1782–1852)

It makes a lot of sense, but haven't we been there before (Example 3) and found that conventional rounding is *not* an apportionment method? Yes, we have tried conventional rounding of the *standard quotas* and found it didn't work, but we haven't considered this approach using *modified quotas*. Now that we know that we can use modified divisors to manipulate the quotas, it is always possible to find a suitable divisor that will make conventional rounding work. This is the idea behind Webster's method (also known as the *Webster-Willcox method* and the *method of major fractions*). Daniel Webster first proposed this method in 1832 (for details on the use of Webster's method to apportion the U.S. House of Representatives see the section "A Brief History of Apportionment in the United States." in the Conclusion).

■ WEBSTER'S METHOD

- **Step 1.** Find a "suitable" divisor d .
- **Step 2.** Using d as the divisor, compute each state's modified quota (modified quota = state population/ d).
- **Step 3.** Find the apportionments by rounding each modified quota to the nearest integer (conventional rounding).

EXAMPLE 8 PARADOR'S CONGRESS (WEBSTER'S METHOD)

To find a suitable divisor for Webster's method we start with the standard divisor $SD = 50,000$ and compute the *standard quotas* (we already did this more than once, but there they are, once again, in the third column of Table 13). When we round the standard quotas to the nearest integer (fourth column of Table 13) we are unlucky and overshoot our target of 250 seats by just one seat. This means that we must bring the quotas down (by just a little) by using a slightly bigger divisor d . Let's try $d = 50,100$. The modified quotas are now shown in the fifth column of Table 13, and when we round these to the nearest integer (last column of Table 13) we get what we want—a Webster apportionment of Parador's Congress.

State	Population	First guess: $SD = 50,000$		Second guess: $d = 50,100$	
		Standard quota	Nearest integer	Modified quota	Nearest integer
A	1,646,000	32.92	33	32.85	33
B	6,936,000	138.72	139	138.44	138
C	154,000	3.08	3	3.07	3
D	2,091,000	41.82	42	41.74	42
E	685,000	13.70	14	13.67	14
F	988,000	19.76	20	19.72	20
Total	12,500,000		251		250

■ **TABLE 13** Webster's method: Calculations for Example 8 with the standard divisor $SD = 50,000$ and the modified divisor $d = 50,100$

Notice that the change that made the difference was very subtle and happened with state *B*, the largest state. The change of divisors dropped *B*'s standard quota from 138.72 to a new quota of 138.44. Dropping *B*'s decimal part to be under 0.5 made all the difference—now *B*'s quota gets rounded down instead of up!

As with the other divisor methods, 50,100 was just one of a range of numbers that work in this example. In fact, any number between 50,080 and 50,385 will work just as well.

Figure 2 shows a flowchart illustrating how to find a divisor d that works for Webster's method. The main difference when we use trial and error to implement Webster's method as opposed to Jefferson's method (compare Figs. 1 and 2) is the choice of the first guess for d . With Webster's method *the first guess should be the standard divisor*. Often (but not always) rounding the standard quotas to the nearest integer does work out. In such cases we have an easy apportionment using Webster's method. Unfortunately, there are times when the standard divisor SD doesn't quite do the job. In such cases we proceed with the trial-and-error approach until we find a divisor that works.

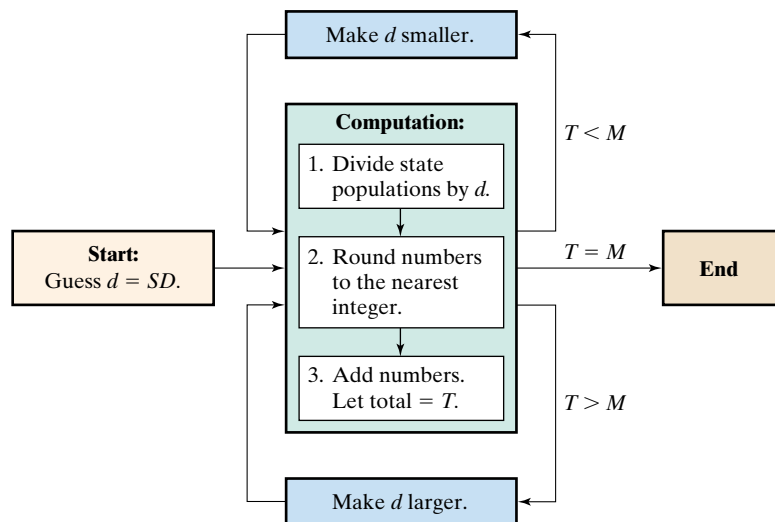


FIGURE 2 Flowchart for trial-and-error implementation of Webster's method.

5

The Huntington-Hill Method

The Dolph Briscoe Center for American History/
The University of Texas at Austin



Edward V. Huntington
(1874–1952)

Until 1940, the method used to apportion the U.S. House of Representatives was chosen by Congress every 10 years, typically for political reasons and with little consideration given to the mathematical subtleties of apportionment or to consistency. Hamilton's method, Jefferson's method, and Webster's method were all used at some time or another. Finally, in 1941 Congress passed, and President Franklin D. Roosevelt signed, "An Act to Provide for Apportioning Representatives in Congress among the Several States by the Equal Proportions Method," generally known as *the 1941 apportionment act*. This act made two important changes in the process of apportioning the seats in the House of Representatives every 10 years: (1) the number of seats is permanently set at $M = 435$, and (2) the method of apportionment is permanently set to be the *Equal Proportions method*, more commonly known as the *Huntington-Hill method*.

If for no other reason than the fact that the Huntington-Hill method is now the official method of apportionment for the U.S. House of Representatives, we will discuss the method in some detail in this section. The Huntington-Hill method is a sophisticated variation of Webster's method, and in many cases it produces exactly the same apportionment as Webster's method. At the same time, the two methods

are *not* equivalent, and in some cases they do produce different apportionments. We will give examples of both of these situations in this section.

But, first things first: The key to understanding the Huntington-Hill method is to understand how the quotas are rounded.

The Geometric Mean and The Huntington-Hill Rounding Rule

Under the Huntington-Hill method, quotas are rounded in a manner that is very similar to but not quite the same as the conventional rounding used in Webster's method. For example, a quota of 2.48 would be rounded down to 2 in Webster's method, but under the Huntington-Hill method 2.48 would be rounded up to 3. How come? Under the Huntington-Hill method, the *cutoff point* for rounding a quota down or up is a tad to the left of 0.5, and it varies with the size of the quota. For example, if the quota is between 2 and 3, the cutoff point is $c = \sqrt{6} \approx 2.4495$ —quotas smaller than c get rounded down, quotas bigger than c get rounded up. Similarly, for a quota between 3 and 4 the cutoff point is $c = \sqrt{12} \approx 3.4641$. To understand these unusual cutoff points for rounding we introduce the two key concepts in this section.

- **Geometric mean** The geometric mean G of two positive numbers a and b is the square root of their product: $G = \sqrt{ab}$. (Note: The geometric mean of a and b always falls somewhere between a and b .)
- **Huntington-Hill rounding rule** Given a quota q , the cutoff for rounding it is given by $c = \sqrt{LU}$ (i.e., the geometric mean of the lower quota L and the upper quota U). In other words, if $q < c$, then round q down to L , and if $q > c$ then round q up to U . (Note: You won't have to worry about the case $q = c$. This can never happen because c is always an irrational number and q is always a rational number.)

The best way to understand how the Huntington-Hill rounding rule works is through an example.

EXAMPLE 9 HUNTINGTON-HILL ROUNDING

Quota is between	Cutoff point
1 and 2	$\sqrt{2} \approx 1.414$
2 and 3	$\sqrt{6} \approx 2.449$
3 and 4	$\sqrt{12} \approx 3.464$
4 and 5	$\sqrt{20} \approx 4.472$
5 and 6	$\sqrt{30} \approx 5.477$
6 and 7	$\sqrt{42} \approx 6.481$
7 and 8	$\sqrt{56} \approx 7.483$
8 and 9	$\sqrt{72} \approx 8.485$
9 and 10	$\sqrt{90} \approx 9.487$

■ **TABLE 14** Cutoff points for Huntington-Hill rounding of quotas between 1 and 10

In each case we will round the quota q up or down using the Huntington-Hill rounding rule. For convenience Table 14 shows cutoff points (and approximations to three decimal places) for all quotas between 1 and 10. [An (*) indicates when the rounding deviates from conventional rounding.]

- $q = 1.51$. This quota gets rounded up to 2, just as it would under Webster's method.
- $q = 1.41$. This quota gets rounded down to 1, just as it would under Webster's method.
- $q = 1.42$. (*) This quota gets rounded up to 2, because it is above the cutoff point $c = \sqrt{2} \approx 1.414$.
- $q = 3.48$. (*) This quota gets rounded up to 4, because it is above the cutoff point $c = \sqrt{12} \approx 3.464$.
- $q = 5.48$. (*) This quota gets rounded up to 6, because it is above the cutoff point $c = \sqrt{30} \approx 5.477$. Notice that here we are just barely above the cutoff point, and we had to go to three decimal places to see the difference between the two numbers.
- $q = 6.48$. This quota gets rounded down to 6, just as under conventional rounding. It is below the cutoff point $c = \sqrt{42} \approx 6.481$.
- $q = 42.49$. Here we need to first calculate the cutoff point c . It is given by the geometric mean of 42 and 43: $c = \sqrt{(42)(43)} = \sqrt{1806} \approx 42.497$. Since q is below the cutoff point, it gets rounded down to 42.

A close look at Table 14 shows that the cutoff points for Huntington-Hill rounding seem to behave quite regularly. The following two facts are pretty useful:

- The decimal part of the cutoff point is always smaller than 0.5, but not by much—it always falls between 0.41 and 0.5.
- The bigger the quota, the closer the decimal part of the cutoff point is to 0.5 (and, thus, the more the rounding rule behaves like conventional rounding).

These two facts reinforce the observation that Huntington-Hill rounding is only a little bit “off center”: Quotas with decimal parts above 0.5 always get rounded up, quotas with decimal parts below 0.41 always get rounded down. There is only a very narrow window to the left of 0.5 where Huntington-Hill rounding is going to be different from conventional rounding.

The Huntington-Hill Method

Once we understand the Huntington-Hill rounding rule, the description of the Huntington-Hill method is straightforward. In fact, it is very similar to the description of Webster’s method.

■ HUNTINGTON-HILL METHOD

- **Step 1.** Find a “suitable” divisor d . [Here a suitable divisor means a divisor that produces an apportionment of exactly M seats when the quotas (populations divided by d) are rounded using the *Huntington-Hill rounding rule*.]
- **Step 2.** Find the apportionment of each state by rounding its quota using the Huntington-Hill rounding rule.

EXAMPLE 10

PARADOR’S CONGRESS (HUNTINGTON-HILL METHOD)

We saw in Example 8 (Parador’s Congress using Webster’s method) that the standard divisor $SD = 50,000$ did not work, but that the modified divisor $d = 50,100$ did. The modified quotas for this divisor are shown in the third column of Table 15 (same quotas as in the fifth column of Table 13). We now need to round these quotas up or down using the Huntington-Hill rounding rules. The cutoff points for rounding are shown in the fourth column of Table 15, and, as is often the case, the rounding turns out to be exactly the same as under conventional rounding. The apportionment under Huntington-Hill method is shown in the last column of Table 15 and is exactly the same apportionment as under Webster’s method.

State	Population	Modified quota	Cutoff point	Round quota to
A	1,646,000	32.85	32.496	33
B	6,936,000	138.44	138.499	138
C	154,000	3.07	3.464	3
D	2,091,000	41.74	41.497	42
E	685,000	13.67	13.491	14
F	988,000	19.72	19.494	20
Total	12,500,000			250

■ **TABLE 15** Huntington-Hill method: Calculations for Example 10 with divisor $d = 50,100$

Example 10 illustrates a fairly common situation: the same divisors that work for Webster’s method work for the Huntington-Hill method. In this case both methods produce exactly the same apportionment.

Our next two examples illustrate situations where the Huntington-Hill method and Webster’s method produce different apportionments. The numbers have been carefully rigged up to illustrate the mathematical subtlety of how this can happen.

EXAMPLE 11 SD WORKS FOR HUNTINGTON-HILL BUT NOT FOR WEBSTER

Parador’s neighbor is the small republic of Guayuru. Guayuru consists of five states (A , B , C , D , and E for short). The populations of the five states are shown in the second column of Table 16. The Guayuru House of Representatives has a total of 40 seats that are to be apportioned to the states using the Huntington-Hill method. Our starting point is to compute the standard divisor and the standard quotas, just as we do with Webster’s method. As luck would have it, the total population of Guayuru is 400,000, which makes the standard divisor $SD = 10,000$ and the calculation of the standard quotas particularly nice. The standard quotas are shown in the third column of Table 16. The fourth column of Table 16 shows the cutoff points for rounding under the Huntington-Hill rounding rules. When the standard quotas are rounded this way the total is 40 (last column of Table 16), and we have an apportionment. Notice that both state A and state E get rounded up—the decimal parts of their quotas are under 0.5 but above the cutoff point. This apportionment is different from the one we would get using Webster’s method (the standard divisor would not work here and we would have to find a different divisor that does—see Exercise 40).

State	Population	Standard quota	Cutoff point	Round quota to
A	34,800	3.48	3.464	4
B	104,800	10.48	10.488	10
C	64,800	6.48	6.481	6
D	140,800	14.08	14.491	14
E	54,800	5.48	5.477	6
Total	400,000	40		40

■ **TABLE 16** Huntington-Hill method: Calculations for Example 11 with the standard divisor $SD = 10,000$

EXAMPLE 12 GUAYURU REVISITED

This example is a small variant of Example 11. There are a few small changes in the populations of the states, but the rest of the story is unchanged. The second column of Table 17 shows the new state populations: the populations of B and C have increased a little, the population of D has decreased. The total is still 400,000, and the standard divisor is still $SD = 10,000$. The standard quotas are shown in the third column of Table 17. When we round the standard quotas using the Huntington-Hill rounding rules (fourth column of Table 17), states B and C now get rounded up and the total number of seats is 42. This does not work, so we have to look for a different divisor. The divisor $d = 10,030$ brings all the quotas down a little. In particular, the quotas of B and E move to the left of their cutoff points and we get the Huntington-Hill apportionment shown in the last column of Table 17. (Note: If we look at the standard quotas in the third column of Table 17 and round them using conventional rounding, we get the apportionment under Webster’s method: A gets 3 seats, B gets 11, C gets 7, D gets 14, and E gets 5.)

State	Population	First guess		Second guess $d = 10,030$	
		Standard quota	Round quota to	Modified quota	Round quota to
A	34,800	3.48	4	3.47	4
B	105,100	10.51	11	10.479	10
C	65,100	6.51	7	6.491	7
D	140,200	14.02	14	13.978	14
E	54,800	5.48	6	5.464	5
Total	400,000	40	42		40

■ **TABLE 17** Huntington-Hill method: Calculations for Example 12 with the standard divisor $SD = 10,000$ and the modified divisor $d = 10,030$

6

The Quota Rule and Apportionment Paradoxes

By and large, each of the apportionment methods we discussed in this chapter has both positive and negative features. Some methods are better than others, but none is flawless. In this section we will discuss the curious flaws that apportionment methods can exhibit and how they are reflected in the methods we studied.

The following discussion can be summarized by three key facts:

1. Except for Hamilton's method, all of the other methods discussed in this chapter violate the *quota rule*.
2. Hamilton's method can produce *apportionment paradoxes*.
3. It is *impossible* for an apportionment method to both *satisfy the quota rule* and not produce *apportionment paradoxes*.

Violating the Quota Rule

- **The Quota rule.** An apportionment method should not apportion to a state less than L seats or more than U seats (where L and U denote, respectively, the lower and upper quotas for the standard quota).

The basic idea behind the quota rule is that the standard quota is the true measure of a fair apportionment, and a fair apportionment method, therefore, should not differ from the standard quota by more than one seat. For example, if a state has a standard quota of 138.72 then it is reasonable to apportion to that state 139 or 138 seats, but never 140 seats or 137 seats.

- **Upper-quota violations.** If an apportionment method apportions to a state more than U seats we say that it *violates the upper quota*. (For example, if the standard quota of a state is 138.72 and the state gets 140 seats, then we have an upper-quota violation.)
- **Lower-quota violations.** If an apportionment method apportions to a state less than L seats we say that it *violates the lower quota*. (For example, if the standard quota of a state is 138.72 and the state gets 137 seats, then we have a lower-quota violation.)

We will now consider how the various methods we learned in this chapter fare with respect to the quota rule.

- **Hamilton's method never violates the quota rule.** The reason for this is that Hamilton's method starts with the standard quotas, rounds all of them down, and then gives surplus seats to some states and not to others. The states that get

a surplus seat end up with their upper quota and the states that don't end up with their lower quota. In either case, the quota rule will not be violated.

- *Jefferson's method can violate the upper quota.* This is illustrated in Example 6 (Parador's Congress), where state B gets an apportionment of 140 seats even though its standard quota is 138.72. Under Jefferson's method all quota violations are *upper-quota* violations [Exercise 71(a)], and these violations tend to happen with the larger states.
- *Adams's method can violate the lower quota.* This is also illustrated with the apportionment of state B in Parador's Congress (Example 7): Under Adams's method B gets an apportionment of 137 seats even though its standard quota is 138.72. Under Adams's method all quota violations are *lower-quota* violations [Exercise 71(b)], and they also tend to happen with the larger states.
- *Webster's method and the Huntington-Hill method can violate the quota rule* (here the violations can be either upper- or lower-quota violations). Quota violations under either method are pretty rare, but they can happen.

Apportionment Paradoxes

Next, we will consider three important paradoxes that sometimes pop up in apportionment problems: the *Alabama paradox*, the *population paradox*, and the *new-states paradox*.

- **Alabama paradox.** This paradox occurs when a state gets apportioned fewer seats simply by virtue of the fact that the total number of seats being apportioned has increased.

We already saw an example of the Alabama paradox in Example 5: Under Hamilton's method a fictitious state called Bama gets apportioned 10 seats when the total number of seats is $M = 200$, but only 9 seats when the total number of seats is increased to $M = 201$.

Of the apportionment methods we discussed in this chapter, only Hamilton's method can produce the quirky arithmetic behind the Alabama paradox: When we increase the number of seats to be apportioned, each state's standard quota goes up, but not by the same amount. As the residues change, some states can move ahead of others in the priority order for the surplus seats. This can result in some state or states losing seats they already had.

[Historical note: The origin of the name *Alabama paradox* can be traced to the events surrounding the 1882 apportionment of the U.S. House of Representatives. As different apportionment bills for the House were being debated, it came to light that in a House with 299 seats, Alabama's apportionment would be 8 seats, whereas in a House with 300 seats, Alabama's apportionment would be 7 seats. The effect of adding that extra seat would be to give one more seat to Texas and one more seat to Illinois. To do this, a seat would have to be taken away from Alabama!]

- **Population paradox.** This paradox occurs when a state gets apportioned fewer seats simply by virtue of the fact that its population has increased. More specifically, the population paradox represents situations in which state X loses a seat to state Y even though X grew in population at a higher rate than Y .

Our next example illustrates how the population paradox can occur under Hamilton's method.

EXAMPLE 13 A TALE OF TWO PLANETS

In the year 2525 the five planets in the Utopia galaxy finally signed a peace treaty and agreed to form an Intergalactic Federation governed by an Intergalactic Congress. This is the story of the two apportionments that broke up the Federation.

Part I. The Apportionment of 2525. The first Intergalactic Congress was apportioned using Hamilton's method, based on the population figures (in billions) shown in the second column of Table 18. (This is the apportionment problem introduced in Example 2.) There were 50 seats apportioned.

Planet	Population	Standard quota	Lower quota	Surplus	Apportionment
Alanos	150	$8.\bar{3}$	8	0	8
Betta	78	$4.\bar{3}$	4	0	4
Conii	173	$9.6\bar{1}$	9	1	10
Dugos	204	$11.\bar{3}$	11	0	11
Ellisium	295	$16.3\bar{8}$	16	1	17
Total	900	50.00	48	2	50

Source: Intergalactic Census Bureau.

■ **TABLE 18** Intergalactic Congress: Apportionment of 2525

Let's go over the calculations. Since the total population of the galaxy is 900 billion, the standard divisor is $SD = \frac{900}{50} = 18$ billion. Dividing the planet populations by this standard divisor gives the standard quotas shown in the third column of Table 18. After the lower quotas are handed out (column 4), there are two surplus seats. The first surplus seat goes to Conii, and the other one to Ellisium. The last column shows the apportionments. (Keep an eye on the apportionments of Betta and Ellisium—they are central to how this story unfolds.)

Part II. The Apportionment of 2535. After 10 years of peace, all was well in the Intergalactic Federation. The Intergalactic Census of 2535 showed only a few changes in the planets' populations—an 8 billion increase in the population of Conii, and a 1 billion increase in the population of Ellisium. The populations of the other planets remained unchanged from 2525. Nonetheless, a new apportionment was required. Table 19 shows the details of the 2535 apportionment under Hamilton's method. Notice that the total population increased to 909 billion, so the standard divisor for this apportionment was $SD = \frac{909}{50} = 18.18$. (You might want to check all the calculations in Table 19 for yourself.)

Planet	Population	Standard quota	Lower quota	Surplus	Apportionment
Alanos	150	8.25	8	0	8
Betta	78	4.29	4	Second	5
Conii	181	9.96	9	First	10
Dugos	204	11.22	11	0	11
Ellisium	296	16.28	16	0	16
Total	909	50.00	48	2	50

Source: Intergalactic Census Bureau.

■ **TABLE 19** Intergalactic Congress: Apportionment of 2535

The one remarkable thing about the 2535 apportionment is that Ellisium *lost a seat* while its *population went up* and Betta *gained that seat* while its *population remained unchanged*! The epilogue to this story is that Ellisium used the 2535 apportionment as an excuse to invade Betta, and the years of peace and prosperity in Utopia were over.

- **New-states paradox.** This paradox occurs when a state's apportionment is affected simply by virtue of the fact that a new state with its fair share of seats has been added to the computation.

[Historical note: In 1907 Oklahoma joined the Union. Prior to Oklahoma becoming a state, there were 386 seats in the House of Representatives. At the time the fair apportionment to Oklahoma was five seats, so the size of the House of Representatives was changed from 386 to 391. The point of adding these five seats was to give Oklahoma its fair share of seats and leave the apportionments of the other states unchanged. However, when the new apportionments were calculated, Maine's apportionment went up from three to four seats and New York's went down from 38 to 37 seats.]

Our last example gives a simple illustration of how the new-states paradox can occur under Hamilton's method. For a change of pace, we will discuss something other than legislatures.

EXAMPLE 14 GARBAGE TIME

District	Homes	Quota	Apportionment
Northtown	10,450	10.45	10
Southtown	89,550	89.55	90
Total	100,000	100.00	100

■ **TABLE 20** Metro garbage truck apportionments

District	Homes	Quota	Apportionment
Northtown	10,450	10.43	11
Southtown	89,550	89.34	89
Newtown	5,250	5.24	5
Total	105,250	105.00	105

■ **TABLE 21** Revised Metro garbage truck apportionments

The Metro Garbage Company has a contract to provide garbage collection and recycling services in two districts of Metropolis, Northtown (with 10,450 homes) and the much larger Southtown (89,550 homes). The company runs 100 garbage trucks, which are apportioned under Hamilton's method according to the number of homes in the district. A quick calculation shows that the standard divisor is $SD = 1000$ homes, a nice, round number which makes the rest of the calculations (shown in Table 20) easy. As a result of the apportionment, 10 garbage trucks are assigned to service Northtown and 90 garbage trucks to service Southtown.

Now imagine that the Metro Garbage Company is bidding to expand its territory by adding the district of Newtown (5250 homes) to its service area. In its bid to the City Council the company promises to buy five additional garbage trucks for the Newtown run so that its service to the other two districts is not affected. But when the new calculations (shown in Table 21) are carried out, there is a surprise: One of the garbage trucks assigned to Southtown has to be reassigned to Northtown! (You should check

these calculations for yourself. Notice that the standard divisor has gone up a little and is now approximately 1002.38.)

Table 22 summarizes the five apportionment methods of this chapter in terms of apportionment paradoxes and the quota rule. Of the five methods, Webster's method and the Huntington-Hill method are in some sense the best because while in theory they can violate the quota rule, in practice this happens very rarely. From the purely mathematical point of view, the ideal apportionment method would be one that does *not* violate the quota rule and does *not* produce any of the apportionment paradoxes. In 1980 two mathematicians—Michel Balinski of the State University of New York at Stony Brook and H. Peyton Young of Johns Hopkins University—were able to prove that such an apportionment method is a mathematical impossibility: *If an apportionment method does not violate the quota rule, then apportionment paradoxes are possible; if an apportionment method does not produce apportionment paradoxes, then violations of the quota rule are possible.* This fact, known as **Balinski and Young's Impossibility Theorem**, forces us to accept the fact that all apportionment methods have one flaw or another.

Method	Quota Rule	Paradoxes		
		Alabama	Population	New-States
Hamilton	No violations	Yes	Yes	Yes
Jefferson	Upper-quota violations	No	No	No
Adams	Lower-quota violations	No	No	No
Webster	Lower- and upper-quota violations	No	No	No
Huntington-Hill	Lower- and upper-quota violations	No	No	No

■ **TABLE 22** Summary of five apportionment methods



Conclusion

Apportionment is a problem of surprising mathematical depth—to say nothing of its political and historical complexities. Many of our Founding Fathers were deeply concerned with the issue, and the names of the apportionment methods we discussed in this chapter bear witness to this fact.

A Brief History of Apportionment in the United States

The apportionment of the U.S. House of Representatives has an intriguing and convoluted history. As mandated by the United States Constitution, apportionments of the House are to take place every 10 years, following each census of the population. The real problem was that the Constitution left the method of apportionment and the number of seats to be apportioned essentially up to Congress. The only two restrictions stated in the Constitution were that (1) “each State shall have at least one Representative,” and (2) “The number of Representatives shall not exceed one for every thirty thousand” (Article I, Section 2).

Following the 1790 Census, and after considerable and heated debate, Congress passed the first “act of apportionment” in 1792. The bill, sponsored by Alexander Hamilton (then Secretary of the Treasury), established a House of Representatives with $M = 120$ seats and apportioned under the method we now call *Hamilton’s method*. In April of 1792, at the urging of Hamilton’s archenemy and then Secretary of State Thomas Jefferson, President George Washington vetoed the bill. (This was the first presidential veto in U.S. history.) Jefferson convinced Washington to support a different apportionment bill, based on a House of Representatives with $M = 105$ seats and apportioned under the method we now call *Jefferson’s method*. Unable to override the president’s veto and facing a damaging political stalemate, Congress finally adopted Jefferson’s apportionment bill. This is how the first House of Representatives came to be constituted.

Jefferson’s method remained in use for five decades, up to and including the apportionment of 1832. The great controversy during the 1832 apportionment debate centered on New York’s apportionment. Under Jefferson’s method, New York would get 40 seats even though its *standard quota* was only 38.59. This apportionment exposed, in a very dramatic way, the critical weakness of Jefferson’s method—it produces *upper-quota violations*. Two alternative apportionment bills were considered during the 1832 apportionment debate, one proposed by John Quincy Adams that would apportion the House using the method we now call *Adams’s method* and a second one, sponsored by Daniel Webster, that would do the same

using the method we now call *Webster's method*. Both of these proposals were defeated and the original apportionment bill passed, but the 1832 apportionment was the last gasp for Jefferson's method.

Webster's method was adopted for the 1842 apportionment, but in 1852 Congress passed a law making Hamilton's method the "official" apportionment method for the House of Representatives. Since it is not unusual for Hamilton's method and Webster's method to produce exactly the same apportionment, an "unofficial" compromise was also adopted in 1852: Choose the number of seats in the House so that the apportionment is the same under either method. This was done again with the apportionment bills of 1852 and 1862.

In 1872, as a result of a power grab among states, an apportionment bill was passed that can only be described as a total mess—it was based on no particular method and produced an apportionment that was inconsistent with both Hamilton's method and Webster's method. The apportionment of 1872 was in violation of both the Constitution (which requires that some method be used) and the 1852 law (which designated Hamilton's method as the method of choice).

In 1876 Rutherford B. Hayes defeated Samuel L. Tilden in one of the most controversial and disputed presidential elections in U.S. history. Hayes won in the Electoral College (despite having lost the popular vote) after Congress awarded him the disputed electoral votes from three southern states—Florida, Louisiana, and South Carolina. One of the many dark sidebars of the 1876 election was that, had the House of Representatives been legally apportioned, Tilden would have been the clear-cut winner in the Electoral College.

In 1882 the *Alabama paradox* first surfaced. In looking at possible apportionments for different House sizes, it was discovered that for $M = 299$ Alabama would get 8 seats, but if the House size were increased to $M = 300$, then Alabama's apportionment would decrease to 7 seats. So how did Congress deal with this disturbing discovery? It essentially glossed it over, choosing a House with $M = 325$ seats, a number for which Hamilton's method and Webster's method would give the same apportionment. The same strategy was adopted in the apportionment bill of 1892.

In 1901 the Alabama paradox finally caught up with Congress. When the Census Bureau presented to Congress tables showing the possible apportionments under Hamilton's method for all House sizes between 350 and 400 seats, it was pointed out that two states—Maine and Colorado—were impacted by the Alabama paradox: For most values of M starting with $M = 350$, Maine would get 4 seats, but for $M = 357, 382, 386, 389$, and 390, Maine's apportionment would go down to 3 seats. Colorado would get 3 seats for all possible values of M except $M = 357$, for which it would only get 2 seats. For $M = 357$, both Maine and Colorado lose, and, coincidentally, this just happened to be the House size that was proposed for the 1901 bill. Faster than you can say "we are being robbed," the debate in Congress escalated into a frenzy of name-calling and accusations, with the end result being that the bill was defeated and Hamilton's method was scratched for good. The final apportionment of 1901 used Webster's method and a House with $M = 386$ seats.

Webster's method remained in use for the apportionments of 1901, 1911, and 1931. (No apportionment bill was passed following the 1920 Census, in direct violation of the Constitution.)

In 1941 Congress passed a law that established a fixed size for the House of Representatives (435 seats) and a permanent method of apportionment known as the *method of equal proportions*, or the *Huntington-Hill method*. The 1941 law (*Public Law 291, H.R. 2665, 55 Stat 761: An Act to Provide for Apportioning Representatives in Congress among the Several States by the Equal Proportions Method*) represented a realization by Congress that politics should be taken out of the apportionment debate and that the apportionment of the House of Representatives should be purely a mathematical issue.

Are apportionment controversies then over? Not a chance. With a fixed-size House, one state's gain has to be another state's loss. In the 1990 apportionment Montana was facing the prospect of losing one of its two seats—seats it had held in the House for 80 years. Not liking the message, Montana tried to kill the messenger. In 1991 Montana filed a lawsuit in federal District Court (*Montana v. United States Department of Commerce*) in which it argued that the Huntington-Hill method is unconstitutional and that either *Adams's method* or *Dean's method* (see Project 1) should be used. (Under either of these methods, Montana would keep its two seats.) A panel of three federal judges ruled by a 2-to-1 vote in favor of Montana. The case then went on appeal to the Supreme Court, which overturned the decision of the lower federal court and upheld the constitutionality of the Huntington-Hill method. So, until either Congress or a different Supreme Court say otherwise, the Huntington-Hill method remains the chosen method for apportioning the U.S. House of Representatives.



KEY CONCEPTS

1 Apportionment Problems and Apportionment Methods

- **apportionment problem:** any problem where M identical, indivisible objects (the “seats”) must be divided among N parties (the “states”) using some proportionality criterion (the “populations”)
- **states:** a metaphor for the parties in an apportionment problem
- **seats:** a metaphor for the identical, indivisible objects being apportioned
- **apportionment method:** given M seats to be apportioned among N states, a method that guarantees a division of the M seats (no more and no less) to the N states based on their populations
- **standard divisor:** the ratio of total population to number of seats being apportioned (i.e., $SD = P/M$)
- **standard quota (quota):** for a state with population p , the ratio p/SD ; it represents what the fair apportionment to the state would be if seats were divisible into fractional parts
- **lower quota:** the largest integer smaller or equal to the quota
- **upper quota:** the smallest integer bigger or equal to the quota

2 Hamilton's Method

- **Hamilton's method:** uses the standard divisor and standard quotas; first gives each state its lower quota, then gives out surplus seats according to the “remainders,” highest remainder first and so on

3 Jefferson's Method

- **Jefferson's method:** a divisor method that produces an apportionment by using modified quotas that are always rounded down
- **divisor (modified divisor):** a number d representing the yardstick for measuring seats in terms of populations (1 seat = d population members)
- **divisor method:** any apportionment method that uses modified divisors and modified quotas for the apportionment calculations
- **modified quota:** a quota obtained using a modified divisor

4 Adams's and Webster's Methods

- **Adams's method:** a divisor method that produces an apportionment by using modified quotas that are always rounded up
- **Webster's method:** a divisor method that produces an apportionment by using modified quotas that are rounded using conventional rounding

5 The Huntington-Hill Method

- **geometric mean:** for two positive numbers, the square root of their product (i.e., geometric mean of a and $b = \sqrt{ab}$.)
- **Huntington-Hill rounding rule:** given a quota q with lower quota L and upper quota U , q is rounded down to L if $q < c$, and rounded up to U if $q > c$
- **Huntington-Hill method:** a divisor method that produces an apportionment by using modified quotas that are rounded according to the Huntington-Hill rounding rule

6 The Quota Rule and Apportionment Paradoxes

- **quota rule:** a state should never be apportioned less than its lower quota or more than its upper quota
- **upper-quota violation:** an apportionment of seats to a state that is more than the state's upper quota
- **lower-quota violation:** an apportionment of seats to a state that is less than the state's lower quota
- **Alabama paradox:** an apportionment paradox where a state may lose seats to another state merely because of an increase in the number of seats being apportioned
- **population paradox:** an apportionment paradox where a state may lose seats to another state merely because its population increased at a higher rate
- **new-states paradox:** an apportionment paradox where a state may lose seats to another state merely because a new state, together with its apportionment of seats, have been added to the apportionment calculations
- **Balinski and Young's Impossibility Theorem:** a perfect apportionment method (no violations of the quota rule and no apportionment paradoxes) is a mathematical impossibility



EXERCISES

WALKING

1 Apportionment Problems and Apportionment Methods

1. The Bandana Republic is a small country consisting of four states: Apure (population 3,310,000), Barinas (population 2,670,000), Carabobo (population 1,330,000), and Dolores (population 690,000). Suppose that there are $M = 160$ seats in the Bandana Congress, to be apportioned among the four states based on their respective populations.

tioned among the four states based on their respective populations.

- (a) Find the standard divisor.
- (b) Find each state's standard quota.

2. The Republic of Wadiya is a small country consisting of four provinces: A (population 4,360,000), B (population 2,280,000), C (population 729,000), and D (population 2,631,000). Suppose that there are $M = 200$ seats in the Wadiya Congress, to be apportioned among the four provinces based on their respective populations.

- Find the standard divisor.
- Find each province's standard quota.

3. The Scotia Metropolitan Area Rapid Transit Service (SMARTS) operates six bus routes (A , B , C , D , E , and F) and 130 buses. The number of buses apportioned to each route is based on the number of passengers riding that route. Table 23 shows the daily average ridership (in thousands) on each route.

- Find the standard divisor.
- Explain what the standard divisor represents in this problem.
- Find the standard quotas.

Route	A	B	C	D	E	F
Ridership	45.3	31.07	20.49	14.16	10.26	8.72

■ TABLE 23

4. The Placerville General Hospital has a nursing staff of 225 nurses working in four shifts: A (7:00 A.M. to 1:00 P.M.), B (1:00 P.M. to 7:00 P.M.), C (7:00 P.M. to 1:00 A.M.), and D (1:00 A.M. to 7:00 A.M.). The number of nurses apportioned to each shift is based on the average number of patients treated in that shift, shown in Table 24.

- Find the standard divisor.
- Explain what the standard divisor represents in this problem.
- Find the standard quotas.

Shift	A	B	C	D
Patients	871	1029	610	190

■ TABLE 24

5. The Republic of Tropicana is a small country consisting of five states (A , B , C , D , and E). The total population of Tropicana is 27.4 million. According to the Tropicana constitution, the seats in the legislature are apportioned to the states according to their populations. Table 25 shows each state's standard quota:

- Find the number of seats in the Tropicana legislature.
- Find the standard divisor.
- Find the population of each state.

State	A	B	C	D	E
Standard quota	41.2	31.9	24.8	22.6	16.5

■ TABLE 25

6. Tasmania State University is made up of five different schools: Agriculture, Business, Education, Humanities, and Science (A , B , E , H , and S for short). The total number of students at TSU is 12,500. The faculty positions at TSU are apportioned to the various schools based on the schools' respective enrollments. Table 26 shows each school's standard quota:

- Find the number of faculty positions at TSU.
- Find the standard divisor.
- Find the number of students enrolled in each school.

School	A	B	E	H	S
Standard quota	32.92	15.24	41.62	21.32	138.90

■ TABLE 26

7. According to the 2010 U.S. Census, 8.14% of the U.S. population lived in Texas. Compute Texas's standard quota in 2010 (rounded to two decimal places). (*Hint:* There are 435 seats in the U.S. House of Representatives.)

8. At the time of the 2010 U.S. Census, New York State had a standard quota of 27.3. Estimate what percent of the U.S. population lived in New York State in 2010. Give your answer to the nearest tenth of a percent. (*Hint:* There are 435 seats in the U.S. House of Representatives.)

9. The Interplanetary Federation of Fraternia consists of six planets: Alpha Kappa, Beta Theta, Chi Omega, Delta Gamma, Epsilon Tau, and Phi Sigma (A , B , C , D , E , and F for short). The federation is governed by the Inter-Fraternia Congress, consisting of 200 seats apportioned among the planets according to their populations. Table 27 gives the planet populations as percentages of the total population of Fraternia:

- Find the standard divisor (expressed as a percent of the total population).
- Find the standard quota for each planet.

Planet	A	B	C	D	E	F
Population percentage	11.37	8.07	38.62	14.98	10.42	16.54

■ TABLE 27

10. The small island nation of Margarita is made up of four islands: Aleta, Bonita, Corona, and Doritos (A , B , C , and D for short). There are 125 seats in the Margarita Congress, which are apportioned among the islands according to their populations. Table 28 gives the island populations as percentages of the total population of Margarita:

- (a) Find the standard divisor (expressed as a percent of the total population).
- (b) Find the standard quota for each island.

Island	A	B	C	D
Population percentage	6.24	26.16	28.48	39.12

■ TABLE 28

2 Hamilton's Method

11. Find the apportionment of the Bandana Republic Congress described in Exercise 1 under Hamilton's method.
12. Find the apportionment of the Wadiya Congress described in Exercise 2 under Hamilton's method.
13. Find the apportionment of the SMARTS buses described in Exercise 3 under Hamilton's method.
14. Find the apportionment of the Placerville General Hospital nurses described in Exercise 4 under Hamilton's method.
15. Find the apportionment of the Republic of Tropicana legislature described in Exercise 5 under Hamilton's method.
16. Find the apportionment of the faculty at Tasmania State University described in Exercise 6 under Hamilton's method.
17. Find the apportionment of the Inter-Fraternia Congress described in Exercise 9 under Hamilton's method.
18. Find the apportionment of the Margarita Congress described in Exercise 10 under Hamilton's method.
19. Happy Rivers County consists of three towns: Dunes, Smithville, and Johnstown. Each year the social workers employed by the county are apportioned among the three towns based on the number of cases in each town over the previous calendar year. The number of cases in each town in 2011 is shown in Table 29.
 - (a) Suppose the total number of social workers employed by the county is $M = 24$. Use Hamilton's method to apportion the social workers to the towns based on the caseloads shown in Table 29.
 - (b) Suppose the total number of social workers employed by the county is $M = 25$. Use Hamilton's method to apportion the social workers to the towns based on the caseloads shown in Table 29.
 - (c) Compare your answers in (a) and (b). What is strange about the two apportionments?

Town	Dunes	Smithville	Johnstown
Number of cases	41	106	253

■ TABLE 29

20. Plainville Hospital has three wings (A, B, and C). The nurses in the hospital are assigned to the three wings based on the number of beds in each wing, shown in Table 30.

- (a) Suppose the total number of nurses working at the hospital is $M = 20$. Use Hamilton's method to apportion the nurses to the wings based on Table 30.
- (b) Suppose an additional nurse is hired at the hospital, bringing the total number of nurses to $M = 21$. Use Hamilton's method to apportion the nurses based on Table 30.
- (c) Compare your answers in (a) and (b). What is strange about the two apportionments?

Wing	A	B	C
Number of beds	154	66	30

■ TABLE 30

3 Jefferson's Method

21. The small nation of Fireland is divided into four counties: Arcadia, Belarmine, Crowley, and Dandia. Fireland uses Jefferson's method to apportion the 100 seats in the Chamber of Deputies among the four counties. Table 31 shows the populations of the four counties after the most recent census.
 - (a) Find the standard divisor and the standard quotas for each county.
 - (b) Determine how many seats would be apportioned if each county was given its lower quota.
 - (c) Determine how many seats would be apportioned if the divisor $d = 197,000$ is used to compute the modified quotas and then all of them are rounded down.
 - (d) Determine how many seats would be apportioned if the divisor $d = 195,000$ is used to compute the modified quotas and then all of them are rounded down.
 - (e) Determine how many seats would be apportioned if the divisor $d = 195,800$ is used to compute the modified quotas and then all of them are rounded down.
 - (f) Determine how many seats would be apportioned if the divisor $d = 196,000$ is used to compute the modified quotas and then all of them are rounded down.
 - (g) Without doing any additional computations, find three different divisors that would work under Jefferson's method.

County	Arcadia	Belarmine	Crowley	Dandia
Population	4,500,000	4,900,000	3,900,000	6,700,000

■ TABLE 31

22. The Republic of Galatia is divided into four provinces: Anline, Brock, Clanwin, and Drundell. Galatia uses Jefferson's method to apportion the 50 seats in its House of Representatives among the four provinces. Table 32 shows the populations of the four provinces (in millions) after the most recent census.

- (a) Find the standard divisor and the standard quotas for each province.
- (b) Determine how many seats would be apportioned if each province was given its lower quota.
- (c) Determine how many seats would be apportioned if the divisor $d = 500,000$ is used to compute the modified quotas and then all of them are rounded down.
- (d) Determine how many seats would be apportioned if the divisor $d = 530,000$ is used to compute the modified quotas and then all of them are rounded down.
- (e) Determine how many seats would be apportioned if the divisor $d = 520,000$ is used to compute the modified quotas and then all of them are rounded down.
- (f) Determine how many seats would be apportioned if the divisor $d = 510,000$ is used to compute the modified quotas and then all of them are rounded down.
- (g) Without doing any additional computations, find three different divisors that would work under Jefferson's method.

Province	Anline	Brock	Clanwin	Drundell
Population (in millions)	5.9	7.8	6.1	6.9

■ TABLE 32

23. Find the apportionment of the Bandana Republic Congress as described in Exercise 1 under Jefferson's method.
24. Find the apportionment of the Wadiya Congress described in Exercise 2 under Jefferson's method.
25. Find the apportionment of the SMARTS buses described in Exercise 3 under Jefferson's method.
26. Find the apportionment of the Placerville General Hospital nurses described in Exercise 4 under Jefferson's method.
27. Find the apportionment of the Republic of Tropicana legislature described in Exercise 5 under Jefferson's method.
28. Find the apportionment of the faculty at Tasmania State University described in Exercise 6 under Jefferson's method.
29. Find the apportionment of the Inter-Fraternia Congress described in Exercise 9 under Jefferson's method. (*Hint*: Express the modified divisors in terms of percents of the total population.)
30. Find the apportionment of the Margarita Congress described in Exercise 10 under Jefferson's method. (*Hint*: Express the modified divisors in terms of percents of the total population.)

4 Adams's and Webster's Methods

31. Find the apportionment of the Republic of Tropicana legislature described in Exercise 5 under Adams's method.
32. Find the apportionment of the faculty at Tasmania State University described in Exercise 6 under Adams's method.

33. Find the apportionment of the Inter-Fraternia Congress described in Exercise 9 under Adams's method. (*Hint*: Express the modified divisors in terms of percents of the total population.)
34. Find the apportionment of the Margarita Congress described in Exercise 10 under Adams's method. (*Hint*: Express the modified divisors in terms of percents of the total population.)
35. Find the apportionment of the Bandana Republic Congress described in Exercise 1 under Webster's method.
36. Find the apportionment of the Wadiya Congress described in Exercise 2 under Webster's method.
37. Find the apportionment of the Republic of Tropicana legislature discussed in Exercise 5 under Webster's method.
38. Find the apportionment of the faculty at Tasmania State University discussed in Exercise 6 under Webster's method.
39. Find the apportionment of the Inter-Fraternia Congress described in Exercise 9 under Webster's method. (*Hint*: Express the modified divisors in terms of percents of the total population.)
40. The small republic of Guayuru (see Example 11) consists of five states (A, B, C, D , and E for short). The populations of the five states are shown in Table 33. Find the apportionment of the $M = 40$ seats in the Guayuru House of Representatives under Webster's method.

State	A	B	C	D	E
Population	34,800	104,800	64,800	140,800	54,800

■ TABLE 33

5 The Huntington-Hill Method

41. Round each number using the Huntington-Hill rounding rules. (*Hint*: $\sqrt{2} \approx 1.414$ is all the information you need. You won't need a calculator or to look up any tables.)
 - (a) 1.5
 - (b) 1.4
 - (c) 1.41
 - (d) 1.42
 - (e) 1.485313
42. Round each number using the Huntington-Hill rounding rules. (*Hint*: $\sqrt{72} \approx 8.48528$ is all the information you need. You won't need a calculator or to look up any tables.)
 - (a) 8.5
 - (b) 8.4
 - (c) 8.483
 - (d) 8.486
43. In the 2010 apportionment of the U.S. House of Representatives, Rhode Island had a standard quota of 1.488879 and a modified quota of 1.485313. How many seats were apportioned to Rhode Island?

44. In the 2010 apportionment of the U.S. House of Representatives, Missouri had a standard quota of 8.458641 and a modified quota of 8.483. How many seats were apportioned to Missouri?

45. A small country consists of five states: A , B , C , D , and E . The standard quotas for each state are given in Table 34.

- (a) Find the number of seats being apportioned.
(b) Find the apportionment under the Huntington-Hill method.

State	A	B	C	D	E
Standard quota	3.52	10.48	1.41	12.51	12.08

■ TABLE 34

46. A small country consists of five states: A , B , C , D , and E . The standard quotas for each state are given in Table 35.

- (a) Find the number of seats being apportioned.
(b) Find the apportionment under the Huntington-Hill method.

State	A	B	C	D	E
Standard quota	25.49	14.52	8.48	30.71	20.8

■ TABLE 35

47. A small country consists of five states: A , B , C , D , and E . The standard quotas for each state are given in Table 36.

- (a) Find the number of seats being apportioned.
(b) Find the apportionment under the Huntington-Hill method.

State	A	B	C	D	E
Standard quota	3.46	10.49	1.42	12.45	12.18

■ TABLE 36

48. A small country consists of five states: A , B , C , D , and E . The standard quotas for each state are given in Table 37.

- (a) Find the number of seats being apportioned.
(b) Find the apportionment under the Huntington-Hill method.

State	A	B	C	D	E
Standard quota	25.496	14.491	8.486	30.449	21.078

■ TABLE 37

49. A country consists of six states, with the state's populations given in Table 38. The number of seats to be apportioned is $M = 200$.

- (a) Find the apportionment under Webster's method.
(b) Find the apportionment under the Huntington-Hill method.
(c) Compare the apportionments found in (a) and (b).

State	A	B	C	D	E	F
Population	344,970	408,700	219,200	587,210	154,920	285,000

■ TABLE 38

50. A country consists of six states, with the state's populations given in Table 39. The number of seats to be apportioned is $M = 200$.

- (a) Find the apportionment under Webster's method.
(b) Find the apportionment under the Huntington-Hill method.
(c) Compare the apportionments found in (a) and (b).

State	A	B	C	D	E	F
Population	344,970	204,950	515,100	84,860	154,960	695,160

■ TABLE 39

6 The Quota Rule and Apportionment Paradoxes

51. Suppose you are taking a multiple-choice quiz on the apportionment chapter. The question is to find how many seats are apportioned to state X under Hamilton's method. You are given all the necessary information to do the computation, including the fact that the standard quota for state X is 35.41. You are presented with four choices: (A) 37, (B) 34, (C) 33, and (D) 36. You can answer this question without doing any work. Which is the correct answer and why?
52. Suppose you are taking a multiple-choice quiz on the apportionment chapter. The question is to find how many seats are apportioned to state Y under Hamilton's method. You are given all the necessary information to do the computation, including the fact that the standard quota for state Y is 78.24. You are presented with four choices: (A) 80, (B) 77, (C) 79, and (D) 81. You can answer this question without doing any work. Which is the correct answer and why?
53. At the time of the 2000 Census, California's standard quota was 52.45. Under Jefferson's method, California would get an apportionment of 55 seats. What does this say about Jefferson's method?
54. At the time of the 2000 Census, California's standard quota was 52.45. Under Adams's method, California would get an apportionment of 50 seats. What does this say about Adams's method?
55. Suppose you are taking a multiple-choice quiz on the apportionment chapter. The question is to find how many seats are apportioned to state X under Jefferson's method. You are given all the necessary information to do the computation,

including the fact that the standard quota for state X is 35.41. You are presented with four choices: (A) 37, (B) 32, (C) 33, and (D) 34. You can answer this question without doing any work. Which is the correct answer and why?

56. Suppose you are taking a multiple-choice quiz on the apportionment chapter. The question is to find how many seats are apportioned to state Y under Adams's method. You are given all the necessary information to do the computation, including the fact that the standard quota for state Y is 78.24. You are presented with four choices: (A) 80, (B) 81, (C) 76, and (D) 82. You can answer this question without doing any work. Which is the correct answer and why?
57. Suppose you are taking a multiple-choice quiz on the apportionment chapter. You are told that the standard quota for state X is 35.41. The question is to determine which of the given answers *can be ruled out as a possible apportionment* to state X under Webster's method. You are presented with five choices: (A) 33, (B) 34, (C) 37, (D) 38, and (E) None of the previous choices can be ruled out. Which is the correct answer and why?
58. Suppose you are taking a multiple-choice quiz on the apportionment chapter. You are told that the standard quota for state Y is 78.24. The question is to determine which of the given answers *can be ruled out as a possible apportionment* to state Y under Webster's method. You are presented with five choices: (A) 77, (B) 76, (C) 80, (D) 81, and (E) None of the previous choices can be ruled out. Which is the correct answer and why?
59. This exercise refers to the apportionment of social workers in Happy Rivers County introduced in Exercise 19. The answers to parts (a) and (b) in Exercise 19 are an illustration of which paradox? Explain. [Obviously, you need to work out parts (a) and (b) of Exercise 19 if you haven't done so yet.]
60. This exercise refers to the apportionment of nurses to wings in Plainsville Hospital introduced in Exercise 20. The answers to parts (a) and (b) in Exercise 20 are an illustration of which paradox? [Obviously, you need to work out parts (a) and (b) of Exercise 20 if you haven't done so yet.]

Exercises 61 and 62 are based on the following story: Mom found an open box of her children's favorite candy bars. She decides to apportion the candy bars among her three youngest children according to the number of minutes each child spent doing homework during the week.

61. (a) Suppose that there were 11 candy bars in the box. Given that Bob did homework for a total of 54 minutes, Peter did homework for a total of 243 minutes, and Ron did homework for a total of 703 minutes, apportion the 11 candy bars among the children using Hamilton's method.
(b) Suppose that before mom hands out the candy bars, the children decide to spend a "little" extra time on homework. Bob puts in an extra 2 minutes (for a total of 56 minutes), Peter an extra 12 minutes (for a total of 255 minutes), and Ron an extra 86 minutes (for a total of 789 minutes). Using these new totals, apportion the 11 candy bars among the children using Hamilton's method.

- (c) The results of (a) and (b) illustrate one of the paradoxes of Hamilton's method. Which one? Explain.

62. (a) Suppose that there were 10 candy bars in the box. Given that Bob did homework for a total of 54 minutes, Peter did homework for a total of 243 minutes, and Ron did homework for a total of 703 minutes, apportion the 10 candy bars among the children using Hamilton's method.
(b) Suppose that just before she hands out the candy bars, mom finds one extra candy bar. Using the same total minutes as in (a), apportion now the 11 candy bars among the children using Hamilton's method.
(c) The results of (a) and (b) illustrate one of the paradoxes of Hamilton's method. Which one? Explain.
63. This exercise comes in two parts. Read Part I and answer (a) and (b), then read Part II and answer (c) and (d).

Part I. The Intergalactic Federation consists of three sovereign planets: Aila, with a population of 5.2 million, Balin, with a population of 15.1 million, and Cona, with a population of 10.6 million. The Intergalactic Parliament has 50 seats that are apportioned among the three planets based on their populations.

- (a) Find the standard divisor in the Intergalactic Parliament.
- (b) Find the apportionment of the 50 seats to the three planets under Hamilton's method.

Part II. Based on the results of a referendum, the federation expands to include a fourth planet, Dent, with a population of 9.5 million. To account for the additional population the number of seats in the Intergalactic Parliament is increased by 15 to a total of 65. [9.5 million individuals represent approximately 15 seats based on the standard divisor found in (a).]

- (c) Find the apportionment of the 65 seats to the four planets using Hamilton's method.
- (d) Which paradox is illustrated by the results of (b) and (c)? Explain.

64. This exercise comes in two parts. Read Part I and answer (a) and (b), then read Part II and answer (c) and (d).

Part I. A catering company contracts to provide catering services to three schools: Alexdale, with 617 students, Bromville, with 1,292 students, and Canley, with 981 students. The 30 food-service workers employed by the catering company are apportioned among the schools based on student enrollments.

- (a) Find the standard divisor, rounded to the nearest integer.
- (b) Find the apportionment of the 30 workers to the three schools under Hamilton's method.

Part II. The catering company gets a contract to service one additional school—Dillwood, with 885 students. To account for the additional students, the company hires 9 additional food-service workers. [885 students represent approximately 9 workers based on the standard divisor found in (a).]

- (c) Find the apportionment of the 39 workers to the four schools under Hamilton's method.
- (d) Which paradox is illustrated by the results of (b) and (c)? Explain.

JOGGING

65. The small island nation of Margarita consists of four islands: Aleta, Bonita, Corona, and Doritos. The state's population of each island is given in Table 40. The number of seats to be apportioned is $M = 100$.

- (a) Find the apportionment under the Huntington-Hill method.
- (b) Describe any possible quota rule violations that occurred under the apportionment in (a).

State	A	B	C	D
Population	86,915	4,325	5,400	3,360

■ TABLE 40

66. Consider an apportionment problem with N states. The populations of the states are given by $p_1, p_2, \dots, p_N$, and the standard quotas are $q_1, q_2, \dots, q_N$, respectively. Describe in words what each of the following quantities represents.

- (a) $q_1 + q_2 + \dots + q_N$
- (b) $\frac{p_1 + p_2 + \dots + p_N}{q_1 + q_2 + \dots + q_N}$
- (c) $\left(\frac{p_N}{p_1 + p_2 + \dots + p_N} \right) \times 100$

67. For an arbitrary state X , let q represent its standard quota and s represent the number of seats apportioned to X under some unspecified apportionment method. Interpret in words the meaning of each of the following mathematical statements.

- (a) $s - q \geq 1$
- (b) $q - s \geq 1$
- (c) $|s - q| \leq 0.5$
- (d) $0.5 < |s - q| < 1$

68. Consider the problem of apportioning $M = 3$ seats between two states, A and B , using Jefferson's method. Let p_A and p_B denote the populations of A and B , respectively. Show that if the apportionment under Jefferson's method gives all three seats to A and none to B , then more than 75% of the country's population must live in state A . (Hint: Show that a Jefferson apportionment of 3 and 0 seats implies that $p_A > 3p_B$.)

69. Consider the problem of apportioning $M = 3$ seats between two states, A and B , using Webster's method. Let p_A and p_B denote the populations of A and B , respectively. Show that if the apportionment under Webster's method gives all three seats to A and none to B , then more than $83\frac{1}{3}\%$ of the country's population must live in state A . (Hint: Show that a Webster apportionment of 3 and 0 seats implies that $p_A > 5p_B$.)

70. Consider the problem of apportioning M seats between two states, A and B . Let q_A and q_B denote the standard quotas of A and B , respectively, and assume that these quotas have decimal parts that are not equal to 0.5. Explain why in this case

- (a) Hamilton's and Webster's methods must give the same apportionment.
- (b) the Alabama or population paradoxes cannot occur under Hamilton's method. [Hint: Use the result of (a).]
- (c) violations of the quota rule cannot occur under Webster's method. [Hint: Use the result of (a).]

71. (a) Explain why, when Jefferson's method is used, any violations of the quota rule must be upper-quota violations.
- (b) Explain why, when Adams's method is used, any violations of the quota rule must be lower-quota violations.
- (c) Explain why, in the case of an apportionment problem with two states, violations of the quota rule cannot occur under either Jefferson's or Adams's method. [Hint: Use the results of (a) and (b).]

72. **Alternate version of Hamilton's method.** Consider the following description of an apportionment method:

- **Step 1.** Find each state's standard quota.
- **Step 2.** Give each state (temporarily) its *upper quota* of seats. (You have now given away more seats than the number of seats available.)
- **Step 3.** Let K denote the number of extra seats you have given away in Step 2. Take away the K extra seats from the K states with the smallest fractional parts in their standard quotas.

Explain why this method produces exactly the same apportionment as Hamilton's method.

Lowndes's Method. Exercises 73 and 74 refer to a variation of Hamilton's method known as Lowndes's method, first proposed in 1822 by South Carolina Representative William Lowndes. The basic difference between Hamilton's and Lowndes's methods is that in Lowndes's method, after each state is assigned the lower quota, the surplus seats are handed out in order of relative fractional parts. (The relative fractional part of a number is the fractional part divided by the integer part. For example, the **relative fractional part** of 41.82 is $\frac{0.82}{41} = 0.02$, and the relative fractional part of 3.08 is $\frac{0.08}{3} = 0.027$. Notice that while 41.82 would have priority over 3.08 under Hamilton's method, 3.08 has priority over 41.82 under Lowndes's method because 0.027 is greater than 0.02.)

73. (a) Find the apportionment of Parador's Congress (Example 3) under Lowndes's method.
- (b) Verify that the resulting apportionment is different from each of the apportionments found under the other methods discussed in the chapter. In particular, list which states do better under Lowndes's method than under Hamilton's method.
74. Consider an apportionment problem with two states, A and B . Suppose that state A has standard quota q_1 and state B

has standard quota q_2 , neither of which is a whole number. (Of course, $q_1 + q_2 = M$ must be a whole number.) Let f_1 represent the fractional part of q_1 and f_2 the fractional part of q_2 .

- (a) Find values q_1 and q_2 such that Lowndes's method and Hamilton's method result in the same apportionment.
- (b) Find values q_1 and q_2 such that Lowndes's method and Hamilton's method result in different apportionments.
- (c) Write an inequality involving q_1 , q_2 , f_1 , and f_2 that would guarantee that Lowndes' method and Hamilton's method result in different apportionments.

RUNNING

75. The Hamilton-Jefferson hybrid method. The Hamilton-Jefferson hybrid method starts by giving each state its lower quota (as per Hamilton's method) and then apportioning the surplus seats using Jefferson's method.

- (a) Use the Hamilton-Jefferson hybrid method to apportion $M = 22$ seats among four states according to the following populations: A (population 18,000), B (population 18,179), C (population 40,950), and D (population 122,871).

- (b) Explain why the Hamilton-Jefferson hybrid method can produce apportionments that are different from both Hamilton and Jefferson apportionments.
- (c) Explain why the Hamilton-Jefferson hybrid method can violate the quota rule.

76. Explain why Jefferson's method cannot produce

- (a) the Alabama paradox
- (b) the new-states paradox

77. Explain why Adams's method cannot produce

- (a) the Alabama paradox
- (b) the new-states paradox

78. Explain why Webster's method cannot produce

- (a) the Alabama paradox
- (b) the new-states paradox



PROJECTS AND PAPERS

1 Dean's Method

Dean's method (also known as the *method of harmonic means*) was first proposed in 1832 by James Dean. [This is a different James Dean. Dean (1776–1849) was a Professor of Astronomy and Mathematics at Dartmouth College and the University of Vermont.] Dean's method is a divisor method similar to Webster's method but with a slightly different set of rules for rounding the quotas. In this project you are to prepare a report on Dean's method. Your report should include (1) a discussion of the *harmonic mean*, including some of its properties and applications; (2) a description of Dean's method; (3) a discussion of how Dean's method can be implemented using trial and error; and (4) a comparison of Dean's method with Webster's method with an example showing that the two methods can produce different apportionments.

2 Apportionment Methods and the 2000 Presidential Election

After tremendous controversy over hanging chads and missed votes, the 2000 presidential election was decided in the United States Supreme Court, with George W. Bush getting the disputed electoral votes from Florida and thus beating Al Gore by a margin of four electoral votes. Ignored in all the controversy was the significant role that the choice of apportionment method plays in a close presidential election. Would things have turned out differently if the House of Representatives had been apportioned under a different method?

In this project, you are asked to analyze and speculate how the election would have turned out had the House of Repre-

sentatives (and thus the Electoral College) been apportioned under (1) Hamilton's method, (2) Jefferson's method, and (3) Webster's method.

Notes: (1) The 2000 Electoral College was based on the 1990 U.S. Census. (Remember to add 2 to the seats in the House to get each state's Electoral College votes!) You can go to www.census.gov/main/www/cen1990.html for the 1990 state population figures. (2) You can use the Apportionment applets in MyMathLab* to do the calculations.

3 Rank Index Implementations of Divisor Methods

From a computational point of view, the divisor methods of Jefferson, Adams, and Webster can be implemented using a priority system (called a **rank index**) that gives away seats one by one according to a table of values specifically constructed for this purpose. Each of the methods uses a slightly different set of rules to construct the rank index for the seats. In this project you are to describe how to construct the rank index for each of the three divisor methods studied in this chapter (Jefferson, Adams, and Webster). You should use a simple example to illustrate the procedure for each method.

*MyMathLab code required.

4 The First Apportionment of the House of Representatives

Table 41 shows the United States population figures from the 1790 U.S. Census.

State	Population
Connecticut	236,841
Delaware	55,540
Georgia	70,835
Kentucky	68,705
Maryland	278,514
Massachusetts	475,327
New Hampshire	141,822
New Jersey	179,570
New York	331,589
North Carolina	353,523
Pennsylvania	432,879
Rhode Island	68,446
South Carolina	206,236
Vermont	85,533
Virginia	630,560
Total	3,615,920

■ TABLE 41

For the first apportionment of 1792, two competing apportionment bills were considered: a bill to apportion 120 seats using Hamilton's method (sponsored by Hamilton), and a bill to apportion 105 seats using Jefferson's method (sponsored by Jefferson). Originally, Congress passed Hamilton's bill, but Washington was persuaded by Jefferson to veto it. Eventually, Congress approved Jefferson's bill, which became the basis for the first mathematical apportionment of the House of Representatives.

This project has three parts.

Part 1. Historical Research. Summarize the arguments presented by Hamilton and Jefferson on behalf of their respective proposals. [Suggested sources: *The Papers of Alexander Hamilton, Vol. XI*, Harold C. Syrett (editor), and *The Works of Thomas Jefferson, Vol. VI*, Paul Leicester Ford (editor)].

Part 2. Mathematics. Calculate the apportionments for the House of Representatives under both proposals.

Part 3. Analysis. It has been argued by some scholars that there was more than mathematical merit behind Jefferson's thinking and Washington's support of it. Looking at the apportionments obtained in Part 2, explain why one could be suspicious of Jefferson and Washington's motives.

(Hint: Jefferson and Washington were both from the same state.)



ANSWERS

WALKING

1 Apportionment Problems and Apportionment Methods

- (a) 50,000
(b) Apure: 66.2; Barinas: 53.4; Carabobo: 26.6; Dolores: 13.8
- (a) 1000
(b) the average ridership per bus each day
(c) A: 45.30; B: 31.07; C: 20.49; D: 14.16; E: 10.26; F: 8.72
- (a) 137 (b) 200,000
(c) A: 8,240,000; B: 6,380,000; C: 4,960,000; D: 4,520,000; E: 3,300,000
- 35.41
- (a) 0.5%
(b) A: 22.74; B: 16.14; C: 77.24; D: 29.96; E: 20.84; F: 33.08

2 Hamilton's Method

- A: 66; B: 53; C: 27; D: 14
- A: 45; B: 31; C: 21; D: 14; E: 10; F: 9
- A: 41; B: 32; C: 25; D: 23; E: 16
- A: 23; B: 16; C: 77; D: 30; E: 21; F: 33
- (a) Dunes: 3; Smithville: 6; Johnstown: 15
(b) Dunes: 2; Smithville: 7; Johnstown: 16
(c) Dunes's apportionment went from 3 social workers to 2 as the total number of social workers increased from 24 to 25.

3 Jefferson's Method

21. (a) $SD = 200,000$; standard quotas are 22.5 (Arcadia), 24.5 (Belarmine), 19.5 (Crowley), 33.5 (Dandia).
 (b) 98 (c) 99 (d) 102 (e) 100 (f) 100
 (g) We know that $d = 195,800$ and $d = 196,000$ work from parts (e) and (f). Any other divisor in between (for example $d = 195,900$) will also work.
23. $A: 67; B: 54; C: 26; D: 13$
 25. $A: 46; B: 31; C: 21; D: 14; E: 10; F: 8$
 27. $A: 41; B: 32; C: 25; D: 23; E: 16$
 29. $A: 22; B: 16; C: 78; D: 30; E: 21; F: 33$

4 Adams's and Webster's Methods

31. $A: 41; B: 32; C: 25; D: 22; E: 17$
 33. $A: 23; B: 16; C: 77; D: 30; E: 21; F: 33$
 35. $A: 66; B: 53; C: 27; D: 14$
37. $A: 41; B: 32; C: 25; D: 23; E: 16$
 39. $A: 23; B: 16; C: 77; D: 30; E: 21; F: 33$

5 The Huntington-Hill Method

41. (a) 2 (b) 1 (c) 1 (d) 2 (e) 2
 43. 2
 45. (a) 40 (b) $A: 4; B: 10; C: 1; D: 13; E: 12$
 47. (a) 40 (b) $A: 3; B: 11; C: 2; D: 12; E: 12$
49. (a) $A: 34; B: 41; C: 22; D: 59; E: 15; F: 29$
 (b) $A: 34; B: 41; C: 22; D: 59; E: 15; F: 29$
 (c) The apportionments are the same under both methods.

6 The Quota Rule and Apportionment Paradoxes

51. The correct answer is (D). Under Hamilton's method there can be no violations of the quota rule. The only possible apportionment to X is 35 or 36.
 53. Upper-quota violations are possible under Jefferson's method.
 55. The correct answer is (A). The other answers represent lower-quota violations, and Jefferson's method can produce only upper-quota violations.
 57. The correct answer is (E). Under Webster's method both lower- and upper-quota violations are possible.
 59. The Alabama paradox (the apportionment to Dunes went from 3 social workers to 2 for no other reason than the fact that the county hired an additional social worker.)
61. (a) Bob: 0; Peter: 3; Ron: 8
 (b) Bob: 1; Peter: 2; Ron: 8
 (c) The population paradox. For studying 3.70% more, Bob gets a piece of candy. However, Peter, who studies 4.94% more, has to give up a piece.
63. (a) $SD = 618,000$
 (b) Aila: 8; Balin: 25; Cona: 17
 (c) Aila: 9; Balin: 24; Cona: 17; Dent: 15
 (d) The new-states paradox: the addition of Dent (and the addition of the exact number of seats that Dent is entitled to) still impacts the other apportionments: Balin has to give up one seat to Aila.

JOGGING

65. (a) $A: 88; B: 4; C: 5; D: 3$
 (b) An upper-quota violation occurs with A 's apportionment.
67. (a) an upper-quota violation by state X
 (b) a lower-quota violation by state X
 (c) The number of seats that state X has received is within 0.5 of its standard quota.
 (d) The number of seats that state X has received satisfies the quota rule but is not the result of conventional rounding of the standard quota.
69. If the modified divisor is d , then state A has modified quota $p_A/d \geq 2.5$. Also, state B has modified quota $p_B/d < 0.5$. So, $p_A \geq 2.5d > 5p_B$. That is, more than five-sixths of the population lives in state A .
71. (a) In Jefferson's method the modified quotas are larger than the standard quotas, so rounding down will give each state at least the integer part of the standard quota for that state.
- (b) In Adams's method the modified quotas are smaller than the standard quota, so rounding up will give each state at most one more than the integer part of the standard quota for that state.
- (c) If there are only two states, an upper-quota violation for one state results in a lower-quota violation for the other state. Neither Jefferson's nor Adams's method can have both upper- and lower-quota violations. So, when there are only two states, neither method can violate the quota rule.
73. (a) $A: 33; B: 138; C: 4; D: 41; E: 14; F: 20$
 (b) State D is apportioned 41 seats under Lowndes's method (all the methods discussed in the chapter apportion 42 seats to state D). States C and E do better under Lowndes's method than under Hamilton's method.



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